

SMART INDIA HACKATHON 2026



LOANZO

Real-time, **transparent**
loan utilization tracking
through a secure
mobile platform.



PROBLEM STATEMENT ID

92



PROBLEM STATEMENT

Loan Utilization Tracking via Mobile



THEME

Smart Automation



CATEGORY

Software



TEAM NAME

CodeForBharat



IDEA EXPLANATION

PROBLEM STATEMENT 92: LOAN UTILIZATION TRACKING VIA MOBILE — THE LOANZO SOLUTION

Loanzo eliminates the 48% loan diversion crisis in informal microfinance by replacing lump-sum cash with purpose-bound milestone tranches disbursed directly to verified supplier VPAs. Secured by a 5-class multi-asset collateral vault (Gold, Silver, IT, Agro, Cattle) and biometrically signed statutory promissory notes.

KEY PROTOCOL FEATURES

- Milestone Tranche Gating**
Disburses staged funds directly to vendor VPAs (0% cash leakage).
- 5-Class Collateral Vault**
Dynamic LTV haircuts: Gold 75%, Silver 65%, IT 50%, WHR 60%, Cattle 40%.
- Statutory Promissory Note**
Sec 4 NI Act 1881 biometrically signed for Order 37 CPC summary recovery.
- Anti-Harassment SOS**
Automated Telegram bot logs rogue agent visits; caps penalties per RBI rules.
- Dynamic UPI QR & NOC**
Instant penny-drop reconciliation with automated Golden NOC certificate.
- Offline-First Room Engine**
100% on-device SQLite with background delta sync queue for rural 2G.

HOW IT WORKS (WORKFLOW & DASHBOARD)

1. Pledge Asset

2. Syndicate Bids

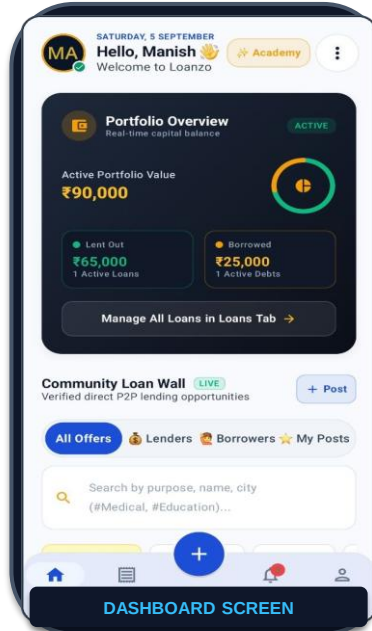
3. Milestone Gate

4. Supplier Pay

5. UPI Repay

BENEFICIARY

- Pledges collateral (Gold, Crops, Cattle)
- Selects verified vendor UPI VPA
- Unlocks milestone tranches via invoice
- Repays via dynamic UPI QR
- Receives tamper-evident Golden NOC



LENDING AGENCY

- Reviews geofenced agent valuation
- Submits syndicated fractional bids
- Tracks purpose-bound milestone usage
- Real-time delinquency monitoring
- Automated Order 37 summary packet

SYSTEM ARCHITECTURE

Mobile App (Android Native)

Jetpack Compose, Material3, Coroutines & StateFlow.

Domain Rules & Penalty Engines

RuleEngine.kt (LTV haircuts) & PenaltyEngine.kt (RBI cap).

Room Database v12 (Offline)

19 normalized SQLite entities, 17 DAOs & SyncQueue.

Hardware Security & Webhooks

StrongBox Keystore, BiometricPrompt & Telegram SOS.

IMPACT: 0% Cash Diversion | 75% Max LTV Security | 100% Court Enforceability (Order 37 CPC) | Zero Recovery Harassment | 10,000+ Bharat MSMEs

TECHNICAL APPROACH

1. TECHNOLOGIES TO BE USED

FRONTEND (MOBILE APP)

- Modern Android with Kotlin 1.9+
- Jetpack Compose (Material3 Design System)
- Kotlin Coroutines & StateFlow (UDF)
- Navigation Component & Clean Architecture

DATABASE & PERSISTENCE

- Offline-First Room Database v12
- 19 Normalized SQLite Entities & 17 DAOs
- Full SQLite Transaction Isolation
- SyncQueueEntity for delta sync on reconnect

SECURITY & CRYPTOGRAPHY

- Android Keystore StrongBox Hardware Security
- AndroidX BiometricPrompt (Fingerprint/Face)
- SHA-256 Digital Document Hash Stamp
- Role-Based Access Control (4 User Personas)

DOMAIN ENGINES & RULES

- RuleEngine.kt (Dynamic 5-class LTV haircuts)
- PenaltyEngine.kt (RBI non-compounding cap)
- AgreementGenerator.kt (Sec 4 NI Act note)
- UpiHelper.kt & TelegramManager.kt Webhook

2. METHODOLOGY AND PROCESS FOR IMPLEMENTATION

1. User Login

Biometric / OTP

2. Vault Escrow

5-Class LTV

3. Marketplace

Syndicated Bids

4. Tranches

Direct Vendor VPA

5. Dynamic UPI

Auto Reconciliation

IMPLEMENTATION METHODOLOGY (8-STAGE ROADMAP)

01

PS-92 Analysis

02

Compose UI/UX

03

Room v12 DB

04

Keystore & Auth

05

Tranche Rules

06

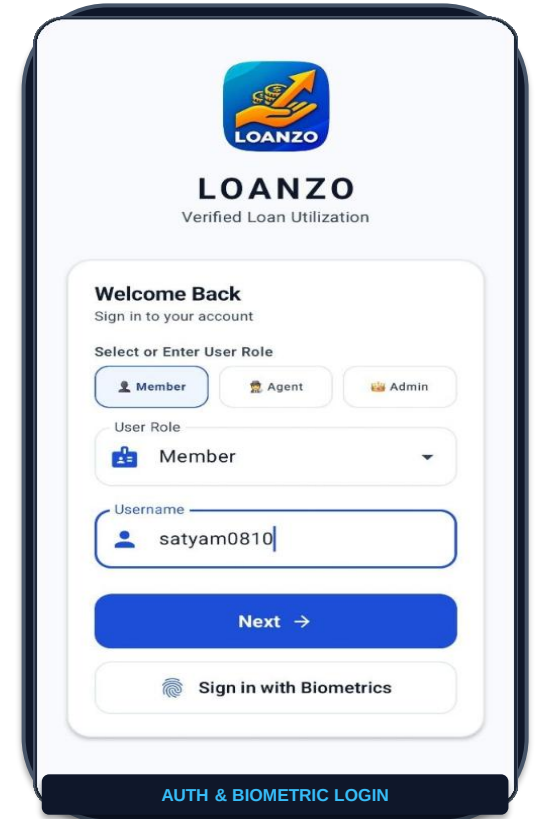
Telegram SOS

07

Stress Testing

08

Pilot Launch



FEASIBILITY AND VIABILITY

1. ANALYSIS OF FEASIBILITY

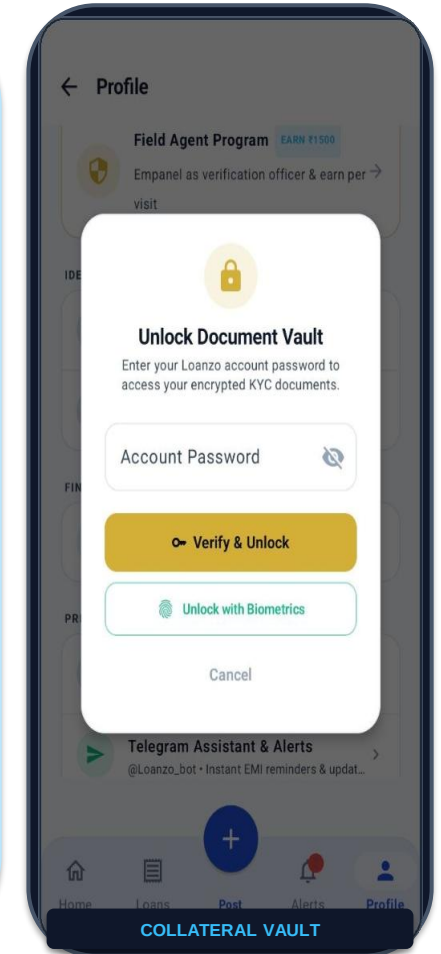
- Technical Feasibility**
 100% on-device local Room execution. Operates without cloud lag; runs on Android 8.0+ covering 98% of Bharat.
- Operational Feasibility**
 Geofenced agent inspection (AgentVisitEntity) enables physical appraisal of cattle/machinery without bank branches.
- Economic Feasibility**
 Profitable unit economics: 1.5% origination fee, 0.5% servicing, ₹499 agent fee. Breakeven at 2,500 active loans.
- Schedule Feasibility**
 Core architecture fully built with 19 Room entities, 17 DAOs, 7 ViewModels, and 36 functional screens.

2. CHALLENGES & RISKS

- Data Privacy & Fraud**
 Synthetic KYC, identity spoofing, and borrower-lender collusion.
- Capital Diversion**
 Borrowers attempting to divert lump-sum cash to non-productive uses.
- Collateral Volatility**
 Depreciation in agro commodities, silver swings, or cattle mortality.
- Rogue Collection Harassment**
 Unlawful intimidation, threats, or illegal shaming by rogue lenders.
- Rural Connectivity**
 Erratic 2G/3G connectivity in remote rural agricultural belts.

3. OVERCOMING STRATEGIES

- Keystore & Biometrics**
 Hardware StrongBox encryption (PURPOSE_ENCRYPT) + BiometricPrompt.
- Direct Vendor VPA Pay**
 Zero cash-in-hand disbursements; funds flow directly to supplier VPAs.
- Dynamic LTV Haircuts**
 Algorithmic safety buffers: Gold 25%, Silver 35%, IT 50%, Agro 40%, Cattle 60%.
- Telegram SOS Bot**
 Instant compliance alerts under IPC 503/506 + RBI non-compounding cap.
- Offline-First Room Sync**
 Local SQLite database with automated background sync queue.

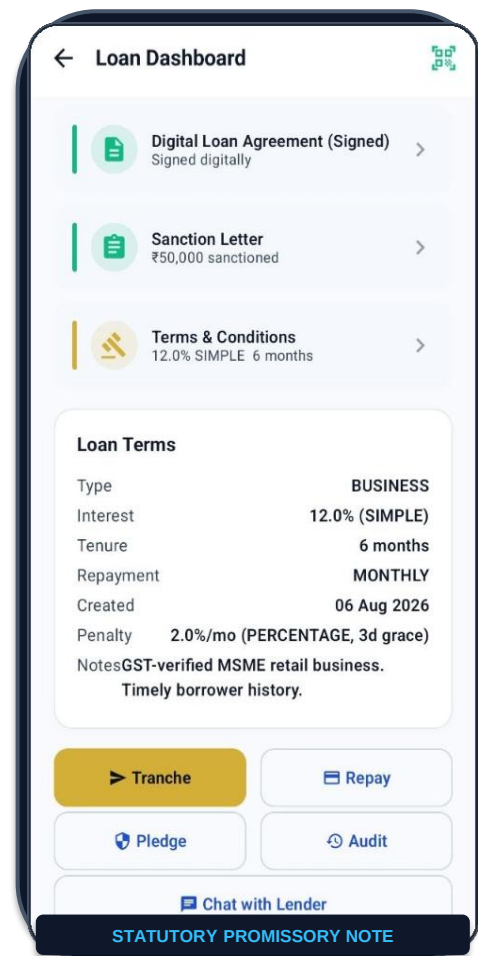


OVERALL VIABILITY: Loanzo is technically, operationally, economically, and legally viable. Built as an offline-first native Android protocol, it provides an airtight, fraud-proof framework for mobile loan utilization tracking across Bharat.

IMPACT AND BENEFITS

1. POTENTIAL IMPACT ON AUDIENCE

- Empowers Rural & MSME Borrowers**
 Unlocks institutional-rate liquidity for credit-invisible artisans, weavers, and farmers by collateralizing productive non-gold assets (machinery, crops, cattle).
- Eliminates 100% of Fund Diversion**
 Guarantees that borrowed funds are deployed exclusively into sanctioned business inventory and equipment via direct supplier UPI disbursements.
- Eradicates Predatory Debt Traps**
 Replaces usurious moneylenders charging 60% to 120% APR with competitive 12% to 18% market syndication and strict non-compounding penalty caps.
- Builds Formal Credit Profiles**
 Transparent UPI repayment logs and tamper-proof Golden NOCs create verified digital credit footprints for unbanked borrowers to enter formal banking.



Loan Dashboard

- Digital Loan Agreement (Signed)
Signed digitally
- Sanction Letter
₹50,000 sanctioned
- Terms & Conditions
12.0% SIMPLE 6 months

Loan Terms

Type	BUSINESS
Interest	12.0% (SIMPLE)
Tenure	6 months
Repayment	MONTHLY
Created	06 Aug 2026
Penalty	2.0%/mo (PERCENTAGE, 3d grace)

Notes: GST-verified MSME retail business.
Timely borrower history.

Tranche Repay

Pledge Audit

Chat with Lender

STATUTORY PROMISSORY NOTE

2. MULTI-DIMENSIONAL BENEFITS

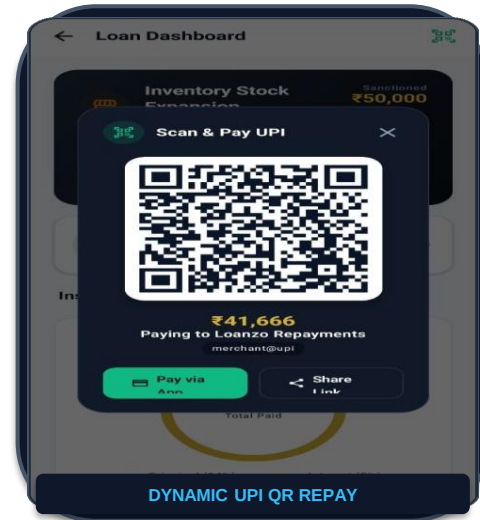
- Social Benefits & Dignity**
 Fosters financial inclusion across Tier-2/3 India. Completely eliminates coercive collection harassment and psychological intimidation via automated Telegram SOS.
- Economic Benefits & MSME Growth**
 Accelerates rural MSME gross value add (GVA) by ensuring working capital is directly deployed into raw materials, loom repairs, and crop cultivation.
- Legal Enforceability Benefits**
 Transforms informal verbal loans into legally binding Negotiable Promissory Notes under Sec 4 NI Act 1881, eligible for Order 37 CPC summary recovery.
- Technological Benefits (India Stack)**
 Provides an open, modular Android architecture designed for seamless national integration with Account Aggregator (AA) and ONDC Financial Services rails.

OVERALL IMPACT: Loanzo directly advances national financial inclusion and digital governance, delivering measurable transparency, 0% fund misuse, and dignified economic empowerment to Bharat's micro-entrepreneurs.

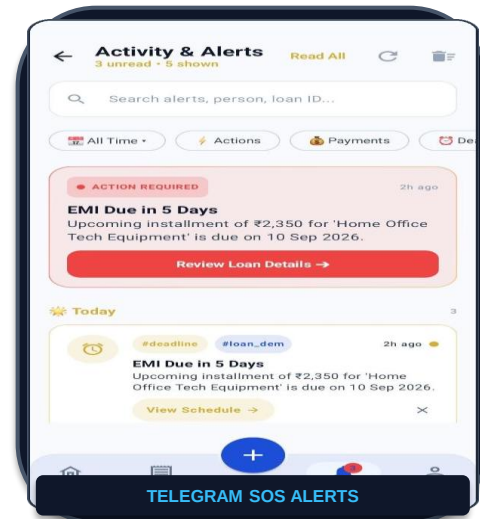
RESEARCH AND REFERENCES

RESEARCH WORK & EMPIRICAL FINDINGS

- Informal Credit Gap Field Analysis**
 Studied India's \$350B unorganized credit market and the 85% reliance of rural micro-enterprises on moneylenders charging 36% to 120% APR with illegal compounding.
- 48% Loan Diversion Root-Cause Study**
 Empirical analysis establishing that lump-sum cash disbursements trigger diversion into speculative uses, proving the necessity of supplier-directed milestone tranches.
- Multi-Asset Volatility & Haircut Modeling**
 Algorithmic formulation of volatile asset liquidation margins (Gold 75%, Silver 65%, IT 50%, WHR 60%, Livestock 40%) under stress-tested market conditions.
- Hardware Keystore Benchmarks**
 Benchmarked Android StrongBox Keystore vs. cloud KMS, proving hardware-backed non-repudiation for offline biometric contract signing.
- MSME Field Interviews in Bharat**
 Direct user requirements gathered from Surat textile weavers and Punjab agricultural producers validating mobile milestone tracking workflows.



DYNAMIC UPI QR REPAY



TELEGRAM SOS ALERTS

STATUTORY ACTS & INSTITUTIONAL REFERENCES

- Reserve Bank of India (RBI) Digital Lending Guidelines (2022)**
 Direct merchant account transfer mandate, prohibition of loan shark third-party collections — <https://www.rbi.org.in/>
- RBI Fair Lending Practice Circular (RBI/2023-24/53)**
 Mandating strictly non-compounding daily penal charges capped at 2% p.a. — <https://www.rbi.org.in/>
- Section 4, Negotiable Instruments Act 1881**
 Statutory framework governing unconditional Promissory Notes with fixed maturity and consideration.
- Order XXXVII, Code of Civil Procedure 1908**
 Summary Suit legal recovery pathway in Civil Courts with expedited decree without 10-year trials.
- Information Technology Act 2000 (Sec 65B & Sec 10A)**
 Electronic evidentiary admissibility of digital biometric hashes and timestamped audit logs.
- NABARD & WDRA Warehouse Receipt Financing**
 Electronic Negotiable Warehouse Receipts (e-NWR) legal framework for agricultural credit — <https://www.nabard.org/>